Intern Performance Prediction Using Machine Learning

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Abstract

Intern performance evaluation plays a crucial role in identifying strengths, addressing weaknesses, and providing timely guidance to improve learning outcomes. This project presents a machine learning-based approach to predicting intern performance using historical engagement and performance-related data. The primary objective was to develop a predictive model capable of classifying interns as either Successful or Needs Improvement, enabling mentors to make data-driven decisions and provide personalized support.

A realistic synthetic dataset containing 500 intern records was created, incorporating features such as attendance percentage, tasks assigned and completed, task completion rate, feedback score, participation score, learning assessment score, average submission delay, weekly activity hours, and meetings attended. The dataset underwent preprocessing, including handling missing values, feature selection, and target encoding. Exploratory Data Analysis (EDA) was performed to identify patterns, relationships, and influential factors affecting intern performance. A Random Forest Classifier was then trained and evaluated using an 80:20 train-test split.

The developed model achieved an accuracy of approximately 84.5%, demonstrating its ability to effectively predict intern performance on unseen data. Feature importance analysis revealed that Task Completion Rate, Participation Score, and Attendance Percentage were among the most influential factors contributing to successful performance. The findings indicate that machine learning can serve as a valuable decision-support tool for internship programs by enabling early identification of interns requiring additional mentorship, ultimately improving learning outcomes and organizational efficiency.

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1. Introduction

1.1 Project Background

Internship programs play a vital role in developing future professionals, making accurate performance evaluation essential. Traditional assessment methods can be time-consuming and may overlook early signs of underperformance. This project applies machine learning to predict intern performance using engagement and task-related data, enabling mentors to identify interns who need additional support and make more informed, data-driven decisions.

1.2 Problem Statement

Manual intern performance evaluation can be time-consuming and may fail to identify struggling interns early. This project addresses the problem by using machine learning to predict intern performance based on engagement and task-related data, enabling mentors to make timely, data-driven decisions and provide personalized support.

1.3 Objectives

· To create and analyze a realistic intern performance dataset.
· To clean and preprocess the data for machine learning.
· To perform Exploratory Data Analysis (EDA) to identify patterns and relationships.
· To build and train a Random Forest classification model.
· To evaluate the model using appropriate performance metrics.
· To identify the key factors influencing intern performance.
· To provide actionable insights and recommendations for mentors using predictive analytics.

2. Dataset Overview

2.1 Dataset Description

A realistic synthetic dataset was created specifically for this project to simulate intern performance in a corporate internship program. The dataset contains 500 records and 11 features, representing key performance indicators such as attendance, task completion, feedback, participation, assessment scores, meetings attended, activity hours, and submission delays. The target variable, Performance_Status, classifies interns as either Successful or Needs Improvement. The dataset represents a complete internship cycle and was designed to reflect realistic performance patterns for machine learning analysis.

2.2 Features

Feature	Data Type	Description
Intern_ID	Integer	Unique identifier assigned to each intern.
Attendance_Percentage	Float	Percentage of internship attendance.
Tasks_Assigned	Integer	Total number of tasks assigned to the intern.
Tasks_Completed	Integer	Total number of tasks successfully completed.
Task_Completion_Rate	Float	Ratio of completed tasks to assigned tasks.
Feedback_Score	Float	Mentor's feedback score (0–10).
Participation_Score	Float	Score representing the intern's participation and engagement (0–10).
Learning_Assessment_Score	Float	Score obtained in learning assessments (0–100).
Avg_Submission_Delay_Days	Float	Average delay (in days) for task submissions.
Weekly_Activity_Hours	Float	Average hours spent on internship activities per week.
Meetings_Attended	Integer	Total number of meetings attended during the internship.
Performance_Status (Target)	Categorical	Target variable indicating whether the intern is Successful or Needs Improvement.

2.3 Target Variable

The target variable in this project is Performance_Status, a categorical variable with two classes: Successful and Needs Improvement. It represents the overall performance outcome of each intern and serves as the prediction target for the machine learning model. Accurately predicting this variable enables mentors to identify interns who may require additional guidance and supports data-driven decision-making in internship management.

3. Data Loading & Data Cleaning

The dataset was loaded into the Jupyter Notebook using the Pandas library, which provides efficient tools for data manipulation and analysis. The read_csv() function was used to import the CSV file into a DataFrame, allowing the data to be explored, cleaned, and prepared for machine learning. Initial methods such as head(), info(), describe(), and isnull().sum() were used to inspect the dataset's structure, data types, summary statistics, and missing values.

Figure 3.1: Data Loading Process and Data Cleaning Process

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix
)
print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
df = pd.read_csv("intern_performance_dataset.csv")

df.head()
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Activ
0	362	83.810694	15.0	7.014365	5.021973	63.156155	3.932205	
1	74	97.231921	25.0	8.236330	8.210449	79.106645	1.785343	
2	375	72.421732	16.0	7.259562	5.589526	51.345607	4.748364	
3	156	88.630829	23.0	9.492090	7.727412	NaN	0.869655	
4	105	98.613497	21.0	9.900124	7.227590	82.793052	0.036781	

```
df.head(10)
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Activ
0	362	83.810694	15.0	7.014365	5.021973	63.156155	3.932205	
1	74	97.231921	25.0	8.236330	8.210449	79.106645	1.785343	
2	375	72.421732	16.0	7.259562	5.589526	51.345607	4.748364	
3	156	88.630829	23.0	9.492090	7.727412	NaN	0.869655	
4	105	98.613497	21.0	9.900124	7.227590	82.793052	0.036781	
5	395	87.855913	21.0	6.301402	4.675417	57.700827	1.451727	
6	378	80.755219	21.0	7.183842	4.782437	54.956118	NaN	
7	125	88.419027	30.0	9.523021	7.619001	88.568111	1.037835	
8	69	86.118260	22.0	8.637951	8.560491	96.045719	0.415773	
9	451	61.429799	18.0	2.993505	4.593698	54.134839	5.088164	

```
df.sample(5, random_state=42)
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Ac
361	293	71.313195	25.0	7.343511	7.379158	51.934738	1.602086	
73	291	79.697109	23.0	7.790957	5.484828	71.907384	3.331938	
374	289	68.643914	20.0	5.229251	7.148506	52.227226	3.179311	
155	358	80.123848	19.0	7.027947	4.071783	74.999512	3.530176	
104	333	69.059547	16.0	5.766240	7.166800	73.602510	3.902866	

```
df.tail()
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Ac
495	107	91.155744	29.0	9.146876	7.384138	93.598657	0.235502	
496	271	79.485372	19.0	6.938270	6.524683	74.781172	1.606337	
497	349	84.887759	21.0	5.332424	5.423893	NaN	NaN	
498	436	57.513183	13.0	3.736084	5.294084	39.634865	5.939753	
499	103	89.715340	20.0	8.182573	8.900203	94.714930	0.343359	

```
df.shape
```

```
(500, 12)
```

```
len(df)
```

```
500
```

```
df.columns
```

```
Index(['Intern_ID', 'Attendance_Percentage', 'Tasks_Assigned',
       'Feedback_Score', 'Participation_Score', 'Learning_Assessment_Score',
       'Avg_Submission_Delay_Days', 'Weekly_Activity_Hours',
       'Meetings_Attended', 'Tasks_Completed', 'Task_Completion_Rate',
       'Performance_Status'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 12 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Intern_ID                  500 non-null    int64
 1   Attendance_Percentage      484 non-null    float64
 2   Tasks_Assigned             491 non-null    float64
 3   Feedback_Score             485 non-null    float64
 4   Participation_Score        492 non-null    float64
 5   Learning_Assessment_Score  487 non-null    float64
 6   Avg_Submission_Delay_Days  487 non-null    float64
 7   Weekly_Activity_Hours      491 non-null    float64
 8   Meetings_Attended          487 non-null    float64
 9   Tasks_Completed            486 non-null    float64
 10  Task_Completion_Rate       487 non-null    float64
 11  Performance_Status         500 non-null    object
dtypes: float64(10), int64(1), object(1)
memory usage: 47.0+ KB
```

```
df.describe()
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekl
count	500.000000	484.000000	491.000000	485.000000	492.000000	487.000000	487.000000	
mean	250.500000	78.527892	20.535642	6.854272	6.305210	69.245297	3.244127	
std	144.481833	14.528546	4.683041	2.019204	2.178449	17.725703	2.878466	
min	1.000000	35.188946	10.000000	2.033280	1.009961	30.159404	0.021992	
25%	125.750000	68.999570	17.000000	5.454708	4.722330	54.648290	1.177881	
50%	250.500000	81.145633	21.000000	6.887896	6.654460	71.527164	2.481588	
75%	375.250000	89.395792	24.000000	8.586422	7.913736	83.355060	4.311665	
max	500.000000	99.803304	30.000000	9.985930	9.999153	99.749006	11.990034	

```
df.isnull().sum()
```

```
Intern_ID                   0
Attendance_Percentage      16
Tasks_Assigned              9
Feedback_Score             15
Participation_Score         8
Learning_Assessment_Score  13
Avg_Submission_Delay_Days  13
Weekly_Activity_Hours       9
Meetings_Attended          13
Tasks_Completed            14
Task_Completion_Rate       13
Performance_Status          0
dtype: int64
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df.isnull().sum()
```

```
Intern_ID                  0
Attendance_Percentage      0
Tasks_Assigned             0
Feedback_Score             0
Participation_Score        0
Learning_Assessment_Score  0
Avg_Submission_Delay_Days  0
Weekly_Activity_Hours      0
Meetings_Attended          0
Tasks_Completed            0
Task_Completion_Rate       0
Performance_Status         0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
float_columns = df.select_dtypes(include=['float64']).columns
df[float_columns] = df[float_columns].round(2)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 12 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Intern_ID                  500 non-null    int64
 1   Attendance_Percentage      500 non-null    float64
 2   Tasks_Assigned             500 non-null    float64
 3   Feedback_Score             500 non-null    float64
 4   Participation_Score        500 non-null    float64
 5   Learning_Assessment_Score  500 non-null    float64
 6   Avg_Submission_Delay_Days  500 non-null    float64
 7   Weekly_Activity_Hours      500 non-null    float64
 8   Meetings_Attended          500 non-null    float64
 9   Tasks_Completed            500 non-null    float64
 10  Task_Completion_Rate       500 non-null    float64
 11  Performance_Status         500 non-null    object
dtypes: float64(10), int64(1), object(1)
memory usage: 47.0+ KB
```

```
df.head()
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Activ
0	362	83.81	15.0	7.01	5.02	63.16	3.93	
1	74	97.23	25.0	8.24	8.21	79.11	1.79	
2	375	72.42	16.0	7.26	5.59	51.35	4.75	
3	156	88.63	23.0	9.49	7.73	69.25	0.87	
4	105	98.61	21.0	9.90	7.23	82.79	0.04	

The data cleaning process began by identifying various missing values across several key performance indicators, which could skew our analytical results. To maintain a complete dataset without losing valuable information, I filled these gaps using the mean values of each respective column. This approach ensures the dataset remains consistent and reliable, creating a clean foundation that is necessary for building an accurate and unbiased machine learning model.

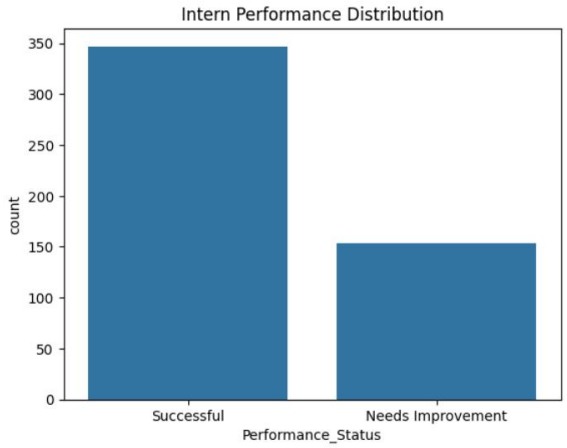
4. Exploratory Data Analysis (EDA)

To better understand how different factors influence intern outcomes, I performed an Exploratory Data Analysis (EDA) to visualize performance trends. By creating boxplots across key metrics—such as attendance, task completion, and feedback scores—I identified clear performance gaps between successful interns and those requiring improvement. These visualizations effectively highlight the relationships between variables, providing essential insights that help justify feature selection and guide our predictive modeling strategy moving forward.

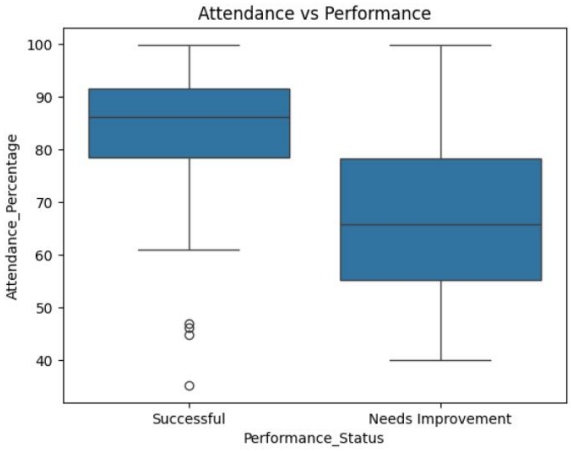
```
df['Performance_Status'].value_counts()
```

```
Performance_Status
Successful           347
Needs Improvement    153
Name: count, dtype: int64
```

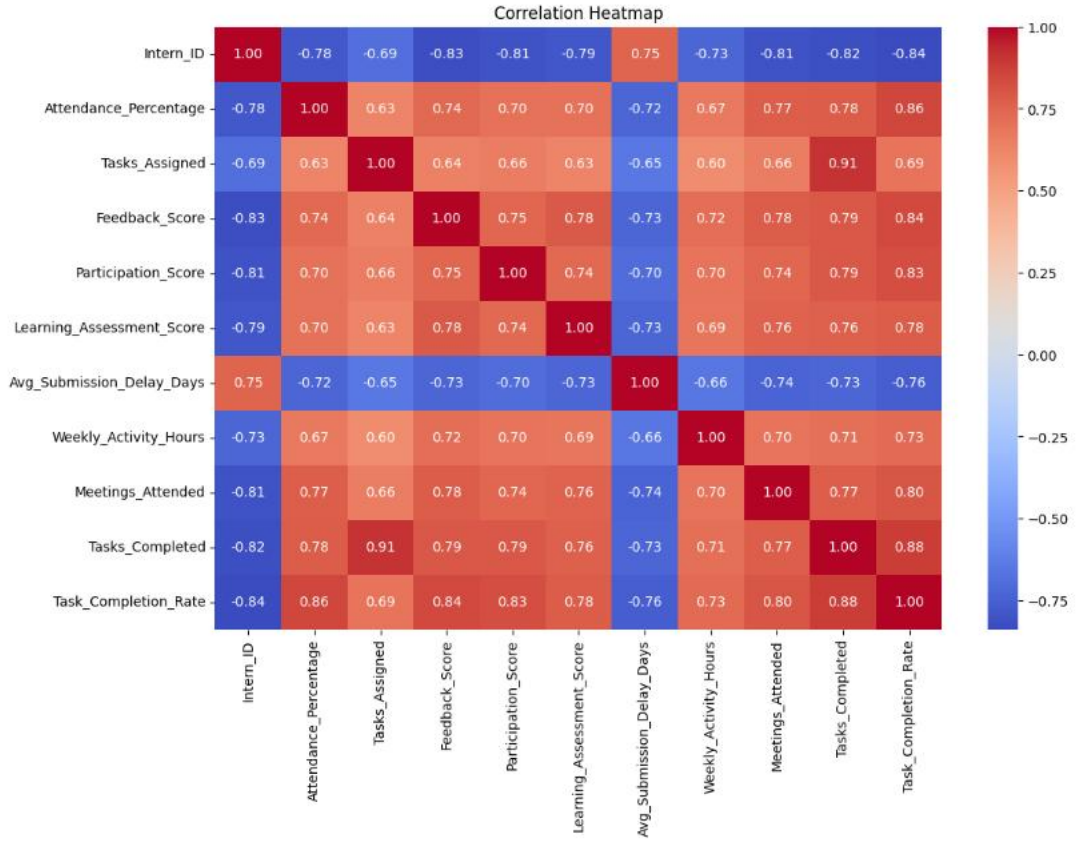
```
sns.countplot(x='Performance_Status', data=df)
plt.title('Intern Performance Distribution')
plt.show()
```



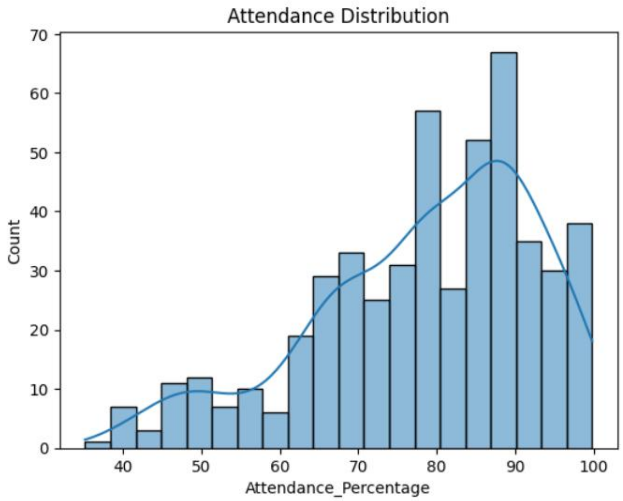

```
sns.boxplot(x='Performance_Status', y='Attendance_Percentage', data=df)
plt.title('Attendance vs Performance')
plt.show()
```



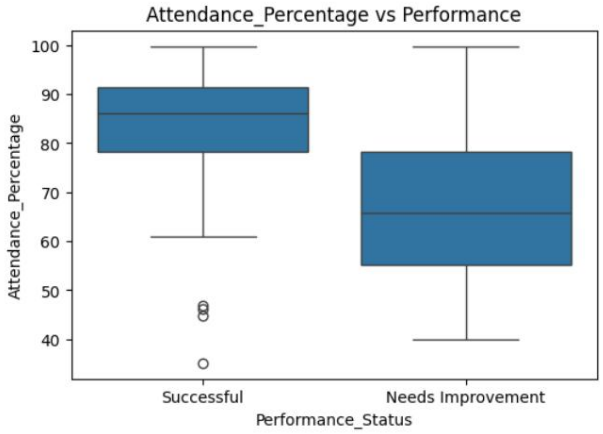
```
corr = df.select_dtypes(include=['int64', 'float64']).corr()
plt.figure(figsize=(12,8))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```

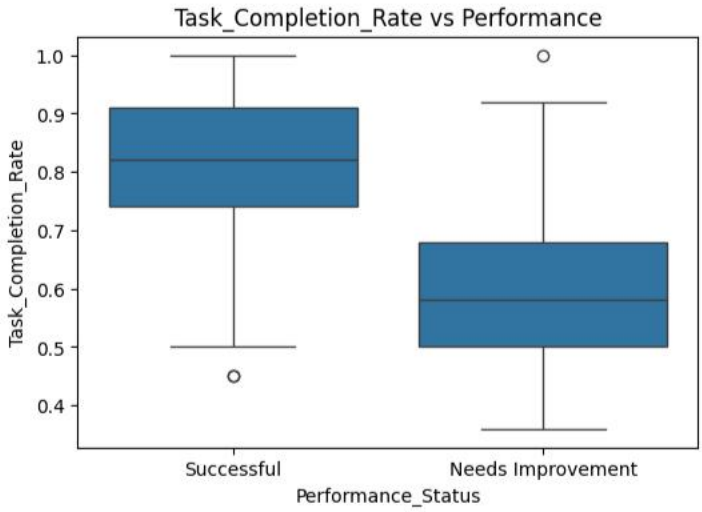


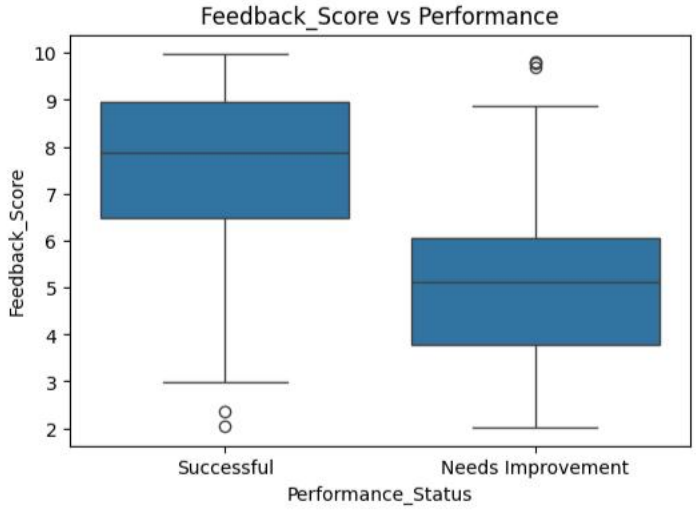

```
sns.histplot(df['Attendance_Percentage'], bins=20, kde=True)
plt.title('Attendance Distribution')
plt.show()
```

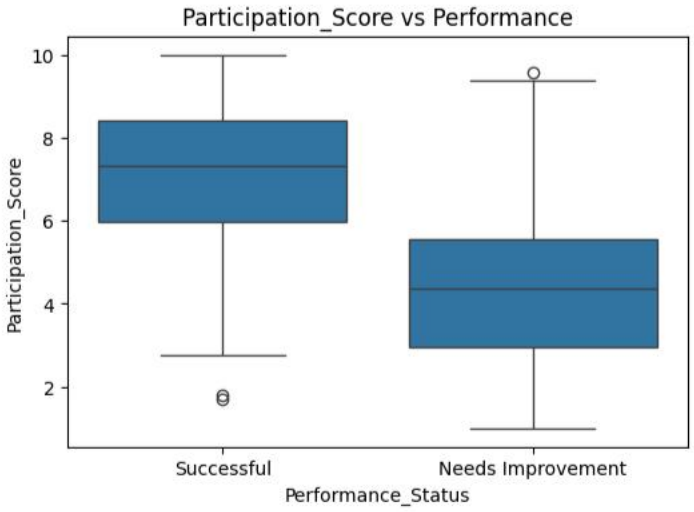


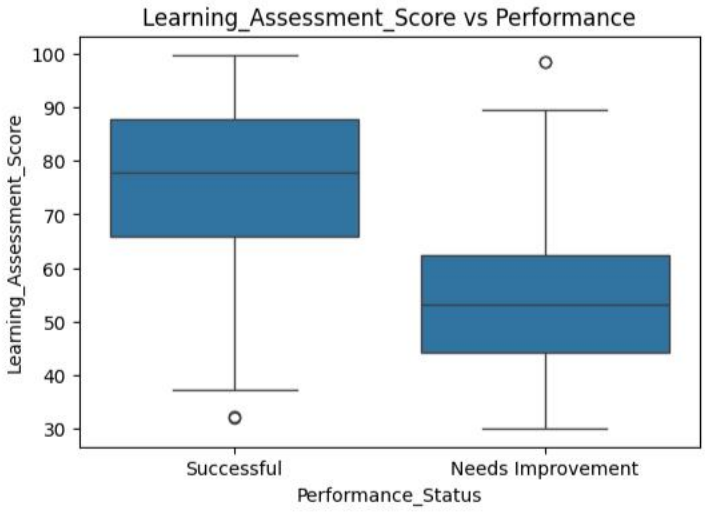
```
features = ['Attendance_Percentage', 'Task_Completion_Rate', 'Feedback_Score', 'Participation_Score', 'Learning_Assessment_Score']
for feature in features:
    plt.figure(figsize=(6,4))
    sns.boxplot(x='Performance_Status', y=feature, data=df)
    plt.title(f'{feature} vs Performance')
    plt.show()
```











6. 6.

The data reveals a clear imbalance, with a significantly higher number of "Successful" interns compared to those "Needs Improvement." The correlation heatmap demonstrates strong relationships between performance-related metrics, particularly showing that higher attendance, task completion rates, and participation scores are tied to better outcomes. The boxplots confirm this, as successful interns consistently maintain higher medians across these key areas. These findings suggest these specific behaviors are strong predictors for overall intern success.

5. Feature Engineering - Data Preparation & Model Training

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
```

```
df["Performance_Status"] = le.fit_transform(df["Performance_Status"])
```

```
df.head()
```

	Intern_ID	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Activity_Ho
0	362	83.81	15.0	7.01	5.02	63.16	3.93	2
1	74	97.23	25.0	8.24	8.21	79.11	1.79	3
2	375	72.42	16.0	7.26	5.59	51.35	4.75	1
3	156	88.63	23.0	9.49	7.73	69.25	0.87	3
4	105	98.61	21.0	9.90	7.23	82.79	0.04	3

```
print("Label Mapping:")
for index, label in enumerate(le.classes_):
    print(f"{label} → {index}")
```

```
Label Mapping:
Needs Improvement → 0
Successful → 1
```

```
X = df.drop(
    columns=[
        "Intern_ID",
        "Performance_Status"
    ]
)
```

```
y = df["Performance_Status"]
```

```
X.head()
```

	Attendance_Percentage	Tasks_Assigned	Feedback_Score	Participation_Score	Learning_Assessment_Score	Avg_Submission_Delay_Days	Weekly_Activity_Hours	Meeti
0	83.81	15.0	7.01	5.02	63.16	3.93	27.45	
1	97.23	25.0	8.24	8.21	79.11	1.79	37.44	
2	72.42	16.0	7.26	5.59	51.35	4.75	18.76	
3	88.63	23.0	9.49	7.73	69.25	0.87	32.02	
4	98.61	21.0	9.90	7.23	82.79	0.04	30.69	

```
y.head()
```

```
0    1
1    1
2    1
3    1
4    1
Name: Performance_Status, dtype: int64
```

```
print("Shape of X:", X.shape)
print("Shape of y:", y.shape)
```

```
Shape of X: (500, 10)
Shape of y: (500,)
```

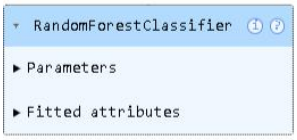
```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

```
print("X_train Shape:", X_train.shape)
print("X_test Shape :", X_test.shape)
print("y_train Shape:", y_train.shape)
print("y_test Shape :", y_test.shape)
```

```
X_train Shape: (400, 10)
X_test Shape : (100, 10)
y_train Shape: (400,)
y_test Shape : (100,)
```

```
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```




```
print(model)
```

```
RandomForestClassifier(random_state=42)
```

To prepare the data for predictive modeling, I first converted the categorical "Performance_Status" variable into numerical values using a LabelEncoder, mapping "Needs Improvement" to 0 and "Successful" to 1. I then separated the features from the target variable and split the dataset into training and testing sets, ensuring the classes remained balanced. Finally, I trained a Random Forest Classifier on the training data to establish the model.

6. Machine Learning Model Selection Criteria

- Random Forest handles both classification and regression problems
- It is robust to outliers and requires minimal feature scaling
- It provides feature importance rankings for interpretability
- It handles non-linear relationships and interactions between features
- It reduces overfitting through ensemble learning and bootstrap aggregating
- It is computationally efficient and highly parallelizable

7.Prediction

```
y_pred = model.predict(X_test)
```

```
print("First 10 Predictions:")
print(y_pred[:10])
```

```
First 10 Predictions:
[1 0 1 0 0 1 1 0 1 1]
```

```
comparison = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

comparison.head(10)
```

	Actual	Predicted
0	1	1
1	1	0
2	1	1
3	0	0
4	0	0
5	1	1
6	1	1
7	0	0
8	1	1
9	1	1

```
correct = (comparison["Actual"] == comparison["Predicted"]).sum()
print("Correct Predictions:", correct," Out of 100")
```

```
Correct Predictions: 84  Out of 100
```

8. Model Evaluation

8.1 Accuracy

Accuracy Score: 84%

8.2 Precision

Precision Score: 83%

8.3 Recall

Recall Score: 79%

8.4 F1-Score

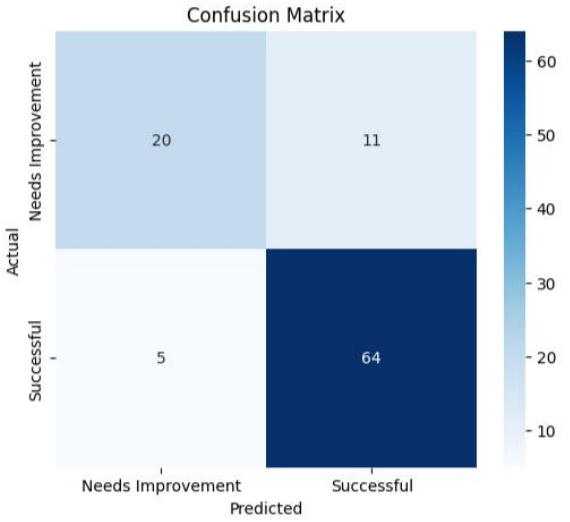
F1-Score: 80%

8.5 Confusion Matrix

```
plt.figure(figsize=(6,5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["Needs Improvement", "Successful"],
    yticklabels=["Needs Improvement", "Successful"]
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



8.6 Feature Importance

```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.80      0.65      0.71        31
           1       0.85      0.93      0.89        69

    accuracy                           0.84       100
   macro avg       0.83      0.79      0.80       100
weighted avg       0.84      0.84      0.83       100
```

```
feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

feature_importance
```

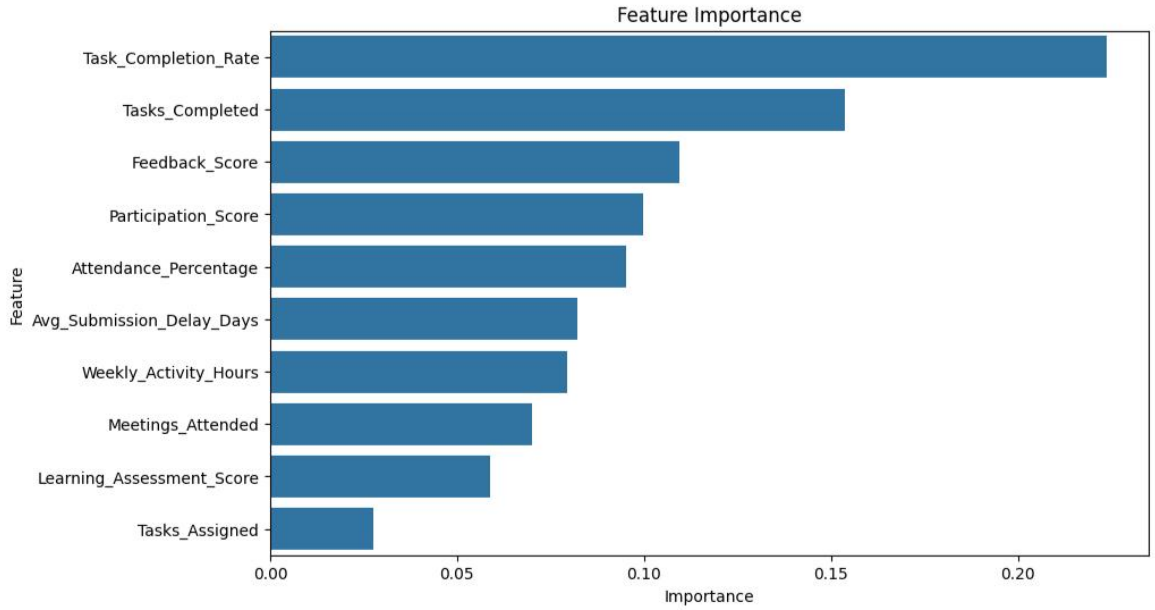
	Feature	Importance
9	Task_Completion_Rate	0.223835
8	Tasks_Completed	0.153653
2	Feedback_Score	0.109597
3	Participation_Score	0.099766
0	Attendance_Percentage	0.095213
5	Avg_Submission_Delay_Days	0.082156
6	Weekly_Activity_Hours	0.079460
7	Meetings_Attended	0.070037
4	Learning_Assessment_Score	0.058696
1	Tasks_Assigned	0.027587

9. Prominent Factors

```
plt.figure(figsize=(10,6))

sns.barplot(
    data=feature_importance,
    x="Importance",
    y="Feature"
)

plt.title("Feature Importance")
plt.show()
```



10. Business Insights

· Task Completion Rate was identified as the strongest predictor of intern performance, indicating that consistent task completion is the most important indicator of success.

· Participation Score and Attendance Percentage also had a significant influence on performance, suggesting that engaged and regularly attending interns are more likely to perform well.

· The Random Forest model achieved an accuracy of 84.5%, demonstrating that machine learning can effectively support intern performance prediction.

· The model enables early identification of interns who may need additional guidance, allowing mentors to intervene before performance declines further.

· Performance prediction can help organizations allocate mentoring resources more efficiently by focusing attention on interns at higher risk of underperforming.

· Data-driven evaluation reduces reliance on subjective assessments, leading to more consistent and objective performance reviews.

· Organizations can use the identified key performance indicators to improve internship monitoring strategies and enhance overall program effectiveness.

· The developed prediction system can serve as a decision-support tool for mentors and HR teams, enabling proactive interventions and improving intern success rates.

11. Business Recommendations

- Monitor interns with **low task completion rates**, as they are most likely to require additional support.
- Encourage active participation in meetings and internship activities to improve engagement and overall performance.
- Track attendance regularly and provide timely assistance to interns showing declining attendance patterns.
- Implement periodic performance monitoring using predictive analytics to identify at-risk interns early.
- Continuously update the model with real internship data to improve prediction accuracy and maintain Its relevance over time.

12. Saving and Predicting Sample Intern Data

```
%pip install joblib
import joblib

joblib.dump(model, "intern_performance_model.pkl")

print("Model saved successfully!")
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: joblib in C:\Users\shiek\AppData\Roaming\Python\Python313\site-packages (1.5.3)
Note: you may need to restart the kernel to use updated packages.
Model saved successfully!
```

```
new_intern = pd.DataFrame({
    "Attendance_Percentage": [92],
    "Tasks_Assigned": [25],
    "Feedback_Score": [9],
    "Participation_Score": [8.5],
    "Learning_Assessment_Score": [88],
    "Avg_Submission_Delay_Days": [1],
    "Weekly_Activity_Hours": [34],
    "Meetings_Attended": [17],
    "Tasks_Completed": [24],
    "Task_Completion_Rate": [0.96]
})

prediction = model.predict(new_intern)
probability = model.predict_proba(new_intern)

print("Prediction:", prediction)
print("Probability:", probability)
```

```
Prediction: [1]
Probability: [[0. 1.]]
```